

A Lending-Lifecycle Fairness Benchmark: Deterministic Ground Truth for Auditing LLM Fairness Interpretation

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Abstract

Large language models are increasingly proposed as fairness auditors, but evaluating them requires ground truth that does not itself come from a model. We release a benchmark over three lending use cases — consumer credit (German Credit), mortgage origination (HMDA Georgia), and peer-to-peer default risk (Lending Club) — in which a deterministic AIF360 pipeline computes every metric and assigns every severity label, and the LLM is scored only on its interpretation of that fixed context. The benchmark ships 36 frozen prompts, a no-API-key replay path, two anti-hallucination detectors, and an enforced cumulative cost cap; full reproduction costs \$0.71. Our results argue against the summary statistic our own benchmark produces. Aggregate severity agreement for `gpt-4o` is 75%, but it decomposes to 33%, 92%, and 100% across the three use cases, and that ordering tracks how discriminating each underlying classifier is rather than model capability. Three of nine attribute pairs changed severity label across four prompt strategies on identical input. Embedding retrieved literature in the prompt cut fabricated citations from 16 to 2 like-for-like. We also report two findings against our own instruments: a refusal detector that flagged six correct terse answers as refusals, and a guardrail that penalised the epistemically correct response to an $n=6$ subgroup. Separately, a metric-orientation error — computing selection rates on the unfavorable label — produced a disparate-impact ratio of 0.0 where the correct value is 0.9931, from a single convention applied to a target named `loan_default`. Following Lee et al. [11], we pair every statistical finding with its decision-context interpretation.

1 Introduction

A fairness audit is a claim about a decision that affects people. A disparate impact ratio of 0.86 is not a finding on its own; it becomes one when attached to who was denied credit, what recourse they had, and what the institution is obliged to check. Lee et al. [11] make this the organising principle of human-centered fair AI: fairness is defined relative to a human decision context, not in metric space. We take that framing seriously as a constraint on what a benchmark can claim.

Recent work has sharpened the practical version of this problem. Zaccour et al. [16] show that quantitative algorithm audits are frequently blocked by data access rather than by method, and Stumpf et al. [15] argue that fairness and correctness assessments need user-centred evaluation rather than metric reporting alone. Kim et al. [10] survey how equity is operationalised across HCI and fairness research and find the construct fragmenting under the practical demands placed on it. Mei et al. [12] demonstrate, in a speech setting, that audit *pipelines* themselves introduce pitfalls that change conclusions. Our contribution sits in that last category: we build the audit infrastructure, then report where the infrastructure misleads.

Table 1: The three use cases. The final row governs how every downstream result reads.

	UC1 German Credit	UC2 HMDA (GA)	UC3 Lending Club
Decision	consumer credit	mortgage origination	P2P default risk
Attributes	sex, age, foreign worker	race, sex, age	gender, income, loan size
Attribute type	protected class	protected class	1 class + 2 proxies
Favorable label	good credit (1)	approved (1)	non-default (0)
Classifier	discriminating	discriminating	~99.5% one class

LLMs are an attractive candidate for the interpretive layer, because generating decision-context narrative from computed metrics is exactly what they are good at. But evaluating them needs ground truth that is not another model’s output. This benchmark supplies that by construction: Python computes the metrics and assigns severity via a fixed rule; the model never recomputes a number, and asserting an unprovided value is recorded as a hallucination, not a result.

Contributions.

1. A three-use-case lending benchmark with deterministic severity ground truth, spanning protected classes and economic proxies, and spanning a discriminating and a near-degenerate classifier.
2. 36 released frozen prompts with a `-dry-run` path that reproduces the entire scoring pipeline with no API key, plus two anti-hallucination detectors and an enforced cumulative cost cap.
3. Two-track evaluation that keeps distinct questions distinct: whether the model interprets metrics reliably (Track Q), and whether it raises the audit-quality ceiling (Track H), never scored against each other.
4. Negative and self-critical results: an aggregate agreement number we argue should not be quoted, and two cases where our own measuring instruments penalise the better answer.

2 Benchmark design

2.1 Use cases

We build on German Credit [8] and HMDA Georgia [5], plus a Lending Club peer-to-peer default task. The three are deliberately not three instances of one setting (Table 1). They differ in the two properties that most change what an audit can conclude: whether the protected attribute is a protected class or an economic proxy, and whether the classifier discriminates enough for disparity to be measurable at all. Lending discrimination is well documented empirically across exactly this span — appearance and gender effects in peer-to-peer markets [6, 13], FinTech-era consumer lending [1], local credit-market conditions [4], and historical records used as training data [14] — which is what makes the lifecycle framing tractable rather than arbitrary.

2.2 Deterministic layer

For each protected attribute we compute disparate impact (DI), demographic parity difference, equal-opportunity difference (EOD), average odds difference (AOD), and the Theil index using AIF360 [2], with Fairlearn [3] mitigations as the reference vocabulary; the canonical interventions the guardrail expects a remediation to name are the standard pre- and post-processing family [7, 9]. A rule-based `classify_severity` maps DI to LOW/MODERATE/HIGH/CRITICAL. This function, not a model, is the ground truth. Intersectional cells are reported with n and suppressed below $n \geq 15$.

Two conventions are pinned by unit tests because both silently changed results during development: the favorable-label orientation (§3.3) and the sign of AOD, whose two compute paths disagreed on privileged-minus-unprivileged ordering while severity narratives quote the sign.

2.3 Frozen prompts and the two tracks

Each (use case $\times$ attribute) pair is rendered into four prompt strategies — **zero_shot**, **chain_of_thought**, **self_critique**, **constrained** — giving $3 \times 3 \times 4 = 36$ prompts. Every prompt embeds the computed metric context, the use-case framing, and retrieved Semantic Scholar evidence, and is stored in full alongside its scored mock response. The benchmark is therefore replayable without re-deriving context, and the released packs double as a no-key regression test.

Track Q asks whether the model is a reliable metric interpreter, and is scored against **classify_severity**. **Track H** asks whether the model raises the audit-quality ceiling, and is judged *only* against an H1–H5 checklist, a mechanical guardrail baseline, and a quality rubric — never against the deterministic numbers, since doing so would re-run Track Q and would penalise an audit for being more careful than a threshold table.

2.4 Anti-hallucination and cost control

Two detectors run on every response. The citation detector flags any reference whose 3-grams do not appear in the retrieved titles; the metric detector flags any asserted numeric value absent from the provided context, excluding canonical thresholds. Both are unit-tested, including a deliberately hallucinated $DI = 0.4321$. Spend is capped cumulatively across invocations with a padded pre-call estimate, so the cap aborts before an overrun rather than after; a persisted ledger makes the cap hold across the separate per-dataset runs.

3 Results

3.1 UC1 German Credit: the marginal view hides the finding

All three marginal DI values clear the four-fifths screen (sex 0.8595, age 0.8212, foreign worker 0.9402). The intersection does not: **under_40 $\times$ female has DI 0.6087** ($n=46$), below the 0.72 CRITICAL threshold that *neither marginal attribute approaches*. Auditing sex and age separately returns three passes and misses the disadvantaged group.

The foreign-worker result is not a finding. Its privileged group has **six records**, under the reporting floor, and the marginal is dominated by the $n=194$ side. *Decision-context reading*: for a loan officer, "this model passes on sex and age" and "this model denies young women at 0.61 the rate of older women" are different operational facts, and only the second is actionable.

3.2 UC2 HMDA: same lesson, different mechanism

Black $\times$ Female ($n=410$) reaches DI 0.7821 against **Asian $\times$ Male** ($n=130$), with a joint disparity exceeding both marginals; both cells are adequately sized. The error profile is diagnostic: race DI 0.8956 is the weakest marginal ratio while $EOD = 0.0025$ and $AOD = 0.0117$ are approximately zero. The model is *not* making differentially worse errors by race — it is reproducing a base-rate difference already present in the labels.

Decision-context reading: this distinction determines who owns the fix. Differential error is a modelling problem; base-rate bias is a policy and data problem upstream of the model, and an equalised-odds post-processor would target a defect this model does not have.

Table 2: Track Q, gpt-4o, 36 calls replayed from the frozen packs. Refusals are split into genuine versus brevity-heuristic artifacts (§3.6).

Cycle	Strategy	Mean score	Severity agr.	Genuine ref.	Halluc. flags
1	zero_shot	76.67	55.6%	0/9	5
2	chain_of_thought	80.00	77.8%	0/9	1
3	self_critique	83.89	88.9%	0/9	1
4	constrained	76.56	77.8%	0/9	1

3.3 UC3 Lending Club: a corrected artifact and an honest null

The target is `loan_default`, so label 1 is the *unfavorable* outcome. Computing selection rates on predicted default gave male 0.0000 versus female 0.0069, hence $DI = 0/0.0069 = 0.0$, reported as CRITICAL. Oriented on the favorable outcome (predicted non-default), the rates are 1.0000 and 0.9931 and $DI = 0.9931$ — near parity. One convention moved the headline metric from maximally alarming to unremarkable.

We stress that nothing here was exotic: a standard toolkit, clean data, a correct metric implementation, and a target variable named for the adverse event. **A practitioner doing everything else right can obtain a maximally wrong conclusion from the orientation default alone.**

The near-parity values are not evidence of fairness. The classifier predicts non-default for $\sim 99.5\%$ of the test set (4 default predictions in 600), so every group looks equal because the model barely discriminates. The honest finding is *insufficient model discrimination to measure disparity*. UC3 also supplies the benchmark’s only economic-proxy contrast: near-parity on `income_level` means something different from near-parity on `gender`, and the pipeline cannot tell them apart.

3.4 Track Q: reliability of metric interpretation

Aggregate severity agreement is 27/36 (75.0%). **We argue against quoting that number.** It decomposes to `german_credit` 33.3%, `hmnda` 91.7%, `lending_club` 100%, and the ordering is not a capability ranking. `lending_club`’s perfect score is the easiest possible case — a near-degenerate classifier makes every attribute a trivially correct near-parity LOW. `german_credit` is hardest: three MODERATE baselines near threshold boundaries where small interpretive differences flip the label. Aggregate agreement is thus a function of how discriminating the audited classifiers are, and a single agreement number for an LLM fairness auditor is not a well-defined quantity.

Errors are asymmetric in the safe direction: 7 over-escalations against 2 under-escalations. Under-escalation is the consequential failure for an audit tool; over-escalation costs review time, not a missed finding.

Three of nine attribute pairs changed severity label across the four strategies on identical input context (Figure 1). Same model, same numbers, different phrasing, different verdict. This — not the agreement rate — is the core reliability result, and it is why we do not recommend letting a model *set* a severity label.

Grounding suppresses fabrication. Comparing like-for-like on the two use cases also covered by an earlier ungrounded run — same model, same attributes, differing only in whether retrieved literature was embedded — fabricated citations fall from **16 to 2** and mean score rises from 69.08 to 76.67 (Figure 2). Retrieval grounding is doing more for factuality here than prompt engineering is.

Provider capability dominates. On the identical harness, Gemini 2.5 Flash refused 19 of 24 cycles with mean score 19.79. A benchmark of this shape measures provider capability at least as much as method, which bounds how much any single-model result should be generalised.

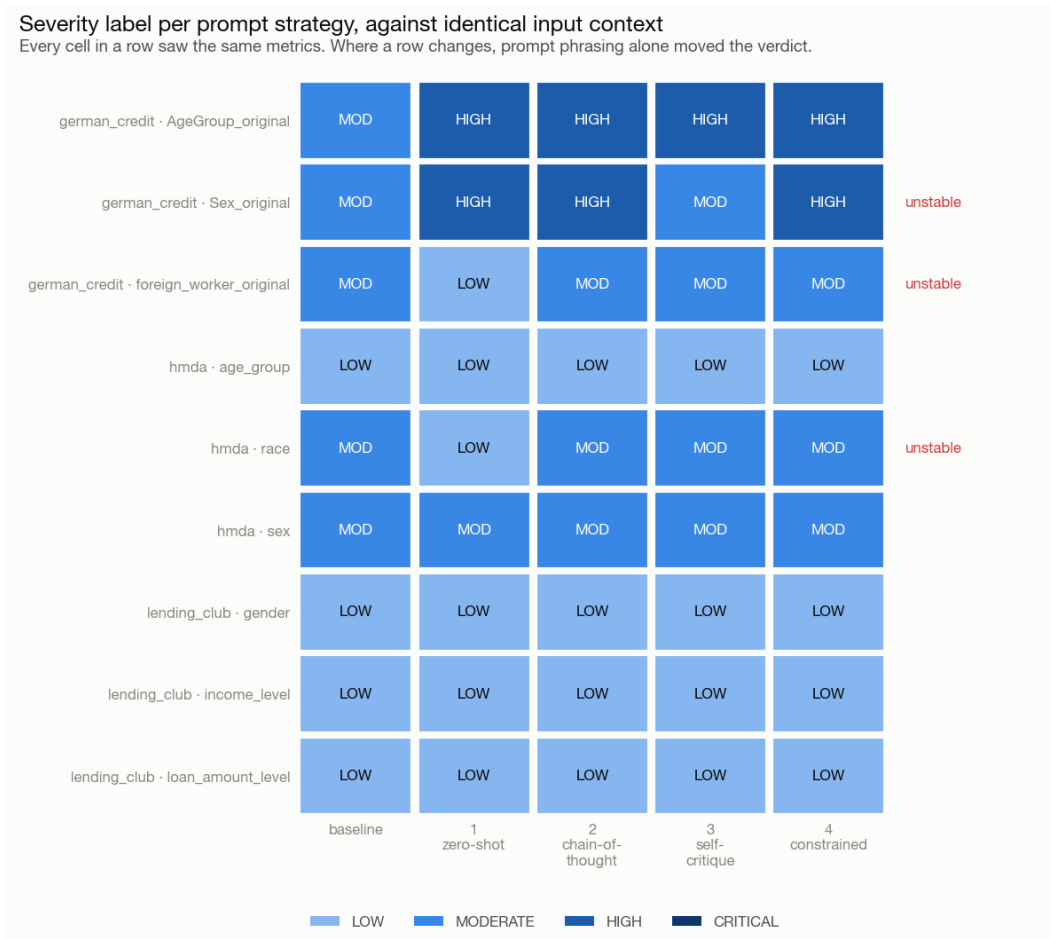


Figure 1: Severity label per prompt strategy against identical input context. Every cell in a row saw the same metrics; where a row changes, prompt phrasing alone moved the verdict.

3.5 Track H: audit quality

Three narrative audits (o3, three self-refinement cycles each) passed 13/15 mechanical guardrail gates and 13/15 H1–H5 criteria, with **27/27 citations verified** against the retrieved evidence pool. A cross-model rubric judge (gpt-4o grading o3) scored 4.8, 4.7, and 5.0 out of 5; we report this as secondary evidence, since an LLM grading an LLM is weaker than a mechanical gate.

Qualitatively, the narrative form supplied things the deterministic template cannot. On foreign worker it reproduced the $n=6$ caveat and quantified it: *"one label change would swing DI to 0.78 (high severity). Thus, statistical power is inadequate."* On UC3 it identified the degenerate classifier unprompted and converted the null into a conditional prediction with a mechanism and a citation. Both are decision-relevant statements with no field in the template to hold them.

The contrast, not a score, is the Track H result: **the deterministic pipeline establishes what is true; the narrative audit establishes what it means for a decision.** They are not competing implementations, and "which is more accurate" is a category error.

Embedding Semantic Scholar evidence cut fabricated citations 8-fold
Same model, same two use cases (german_credit, hmda), 24 vs 24 records.

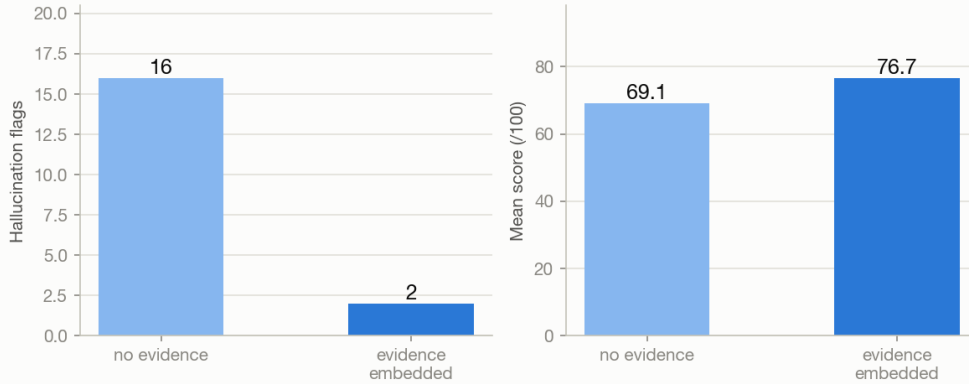


Figure 2: Embedding retrieved Semantic Scholar evidence in the prompt cut fabricated citations roughly eight-fold on the two use cases common to both runs.

3.6 Two findings against our own instruments

The refusal detector conflates brevity with refusal. It flagged 6 of 9 **constrained** responses; **none were genuine refusals**. Any narrative field under 40 characters counts as effectively empty, so `how_to_fix`: "DisparateImpactRemover" — precise and on-spec in 22 characters — registers as a refusal, under a prompt that asked for terse output. Reporting a 66.7% refusal rate would have attributed a measurement artifact to the model. We left the detector unchanged, since it is part of the frozen harness and altering it would invalidate the ungrounded comparison, and report the distinction instead.

A guardrail penalised the correct answer. Our baseline requires every mitigation to name a canonical algorithm. The two attributes that fail are exactly the two whose problem *is* data scarcity — foreign worker ($n=6$) and loan-amount tier ($3.2\times$ imbalance) — where the audit instead proposed acquiring records to a stated floor, Bayesian hierarchical shrinkage, bootstrapped confidence intervals, and manual review. **No post-processor repairs an $n=6$ subgroup**; applying one would manufacture false confidence. The guardrail encodes an assumption that every fairness problem has an algorithmic fix.

Both were written in good faith by people who knew the domain, and both quietly punish the better answer. This is the concrete form Mei et al. [12]’s auditing-pitfalls argument takes in our setting: audit infrastructure carries normative commitments, and they surface as penalties.

4 Discussion

Read through the Lee et al. [11] lens, our results are less about model capability than about where a model can be trusted in a human decision pipeline. The defensible division of labour is the one the benchmark enforces: *deterministic code owns the numbers and the thresholds; the model owns the interpretation; and the interpretation is gated on evidence the model did not invent*. Track Q supports the first clause — severity labels move under paraphrase — and the 16-to-2 fabrication result supports the third.

Two implications for practice. First, marginal-only auditing, which is the common case, systematically misses intersectional harm: in UC1 every marginal passes while the joint group sits at 0.6087. Second, threshold-as-gate reading destroys information. UC2’s race result means label bias,

not model error; UC3’s null means the model is not deciding anything yet. Both are invisible if DI is read as pass/fail — which is how it tends to be used, and what Stumpf et al. [15] and Kim et al. [10] identify as the gap between fairness metrics and the assessments practitioners actually need.

5 Limitations

- **Scale.** Three datasets, one primary model. This is a benchmark design with a worked demonstration, not a large-scale evaluation.
- **Subgroup sizes.** Foreign-worker privileged $n=6$; several HMDA race categories are severely sparse. Cells below $n \geq 15$ are suppressed and n is always reported.
- **UC3 informativeness.** A classifier predicting one class $\sim 99.5\%$ of the time cannot exhibit measurable disparity; UC3 exercises the pipeline, not the fairness of a working model.
- **LLM nondeterminism.** No seed or temperature pinning. The four-cycle design varies prompt strategy, so cycle-to-cycle differences confound strategy with sampling noise; we cannot separate them.
- **API versioning.** gpt-4o and o3 are moving targets. The frozen packs make inputs reproducible; outputs are not.
- **Metric grounding.** All metrics are associational. Root-cause mapping is rule-based inference from metric values and feature correlations, **not causal identification**, and we make no causal claim.
- **No legal claim.** ECOA, Regulation B, and the four-fifths rule appear as decision context. A DI below 0.8 is a screening signal, not a finding of unlawful discrimination.
- **Cost figures.** o3 is priced at the higher historical rate, so its cost is an upper bound rather than a billed amount.
- **Generality.** HMDA is Georgia only; German Credit is 1990s data used as a benchmark, not a current population.

6 Conclusion

We release a lending-lifecycle fairness benchmark whose ground truth is deterministic, whose 36 prompts are frozen and replayable without an API key, and whose full reproduction costs under a dollar. Its most useful outputs are the uncomfortable ones: an aggregate agreement statistic that we show is not a well-defined quantity, severity labels that move under paraphrase, a disparate-impact ratio that read 0.0 instead of 0.9931 because of one orientation default, and two of our own measuring instruments penalising the correct answer. Following Lee et al. [11], we take the last group most seriously: audit infrastructure is not a neutral lens, and a benchmark that cannot detect its own biases is not measuring what it claims to.

Availability. Code, all 36 frozen prompt packs, per-dataset metrics, dataset cards, the Track Q and Track H analyses, and the spend ledger are in the repository. `make test` runs 46 unit tests; `make fairness` regenerates the deterministic layer; `-dry-run` reproduces the full scoring path with no API key.

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